

* Unit - 4 *

Advance technical writing

A technical report is a formal document that describes the process, progress, results or analysis of a technical or scientific work.

It explain what was done, how it was done & what was found.

Purpose of technical report

- To present facts & data clearly
- To analyze technical problems
- To help decision-making
- To document research or project work

Characteristics of Technical Report:-

A good technical report has specific features that makes it clear, effective & professional.

These are explained :-

(i) Shape & Size -

- Usually 8-10 pages or more.
- Proper format, margin, spacing
- length depends on topic & requirement.

(ii) Content -

- Should be easy to understand
- Even a non-technical person should follow it
- Avoid too much complexity.

(3) Facts & Data:-

- Must include accurate facts & figures
- Based on objective & purpose
- No. unnecessary information

(4) Sources:-

Information collected from:-

- Books
- Journals
- Experts
- Surveys

Source must be reliable

(5) Documentation:-

- Proper references & citations
- Acknowledges original sources
- Ensures authenticity

(6) Target Readers -

- Written for a specific audience
- Content should match their knowledge level.

(7) Titles & Subtitles -

- Use headings & subheadings
- Helps in -

- Better organization
- Easy to understand
- Quick navigation

(8) Physical presentation -

- Neat & properly typed / printed
- Use proper binding (prefer hard binding)
- Good presentation improves impact.

(9) Graphics & visuals -

Use graphs, charts, Diagrams, tables.
Makes report easy to understand, more attractive & more authentic.

* Types of Technical Reports

Technical reports can be classified based on their purpose & use. According to

(1) Technical Background Report -

- Provides background info information on a topic.

- Helps readers understand basic concepts.

Ex Report on wind energy or VLSI tech.

(2) Instruction Report -

- Gives step-by-step procedures

- Used in labs & manuals.

Ex Lab experiment procedure.

(3) Feasibility Report :-

- Check whether a project is possible or practical

- Analyzes cost, resources & benefits

Ex Feasibility of a new manufacturing process.

(4) Recommendation Report:-

Suggest the 'best option or decision',
Prepared after evaluation.

Ex: selecting a supplier or technology.

(5) Evaluation Report:-

- Evaluates performance or results

- ~~Helps~~ Helps in decision-making.

Ex: Evaluation of a new syllabus or system.

(6) Primary Research Report:-

- Based on original research or experiment

- Includes data analysis & conclusions

Ex: Research project Report.

(7) Technical specification Report:-

- Describes technical details of a product

- Includes size, material, features, performance

Ex: Specifications of a machine or device.

(8) Proposal - Based Report:-

- Detailed reports prepared for big projects

- Includes feasibility, research & planning

(9) Business Prospectus:-

- Describes a business plan or idea

- Includes market analysis & future plans

Other common types

- Progress Report
- Trip report
- Laboratory report
- Inter-Inspection Report
- Investigation Report

* Structure & format of Technical report:-

A technical report follows a proper structure so that information is clear, organized & easy to understand.

(1) Cover Page (Title Page) -

Contains - Title of Report

- Name of author/organization
- Date
- Submitted to whom.

Gives first impression of the report.

(2) Summary (Abstract) -

- Short overview of the report
- Includes - Purpose
- Method
- Key finding

Helps reader quickly understand the report

(3) Table of Content:-

Lists all sections with pages numbers
Helps in easy navigation.

(4) Introduction:-

Explains:

- Problem/Topic

- Background

- Objectives

Defines scope of report.

(5) Main Body (Core section):-

Detailed explanation of

- Method

- Tools / techniques

- Process

May includes diagrams, flowcharts.

(6) Results:-

Presents findings using:-

- Tables

- Graphs

- Data

(7) Conclusion:-

- Summarization key results

- Gives final outcomes.

(8) Recommendation:-

- Suggestions based on results

- Future action or improvement

(9) References / Bibliography

- List of sources used
- Books, journals, websites

(10) Appendices -

- Extra information (data, charts etc.)
- Supports the main report

* Technical Proposal :-

A technical proposal is a formal document that explains a plan or solution for a project, along with methods, requirements, cost & timeline.

It is a written offer to do some work or solve a problem.

Purpose of Technical Report -

- To propose a solution
- To get approval or funding
- To win a project or contract

Key features -

- Clearly defines the problem & solution
- Includes technical details
- Mentions cost & time schedule
- Written in formal & professional style

Types of Technical Reports Proposal:-

(1) Business Proposal -

For business activities like

- Construction
- Supply of materials
- Service

Ex Proposal for building a bridge.

(2) Research Proposal -

- For conducting research projects
- Submitted to funding agencies

Ex Proposal for AI research project.

(3) Educational Proposal

for starting schools, colleges or course.

Ex Proposal to open a new engineering college.

(4) Accreditation Proposal

For getting quality certification

Evaluates performance to standards.

Ex NAAC or NBA accreditation.

Basic Structure of Technical Proposal:

1. Title
2. Introduction (Problem)
3. Objectives
4. Methodology (How work will be done)
5. Resource required
6. Time schedule
7. Budget / Cost
8. Conclusion

* Technical Articles

A technical article is a written document that ~~para~~ presents technical or scientific information in a clear & structured form for a specific audience.

It explains a technical topic, research or idea in details.

Purpose of Technical Article

- To share knowledge or research
- To inform & educate readers
- To explain technical concepts clearly.

Key features

- Clear & precise language
- Logical structure
- Based on facts & evidence
- Written for specific audience

Types of Technical Article:

(1) Scientific Article -

Based on research & experiments
Uses data & evidence.

Ex Research paper in science journal

(2) Research Paper -

Detailed study of a topic

Includes -

- Data from books, articles, internet
- Personal analysis & ideas.

Ex Engineering research paper.

Structure of Technical Article

1. Title
2. Abstract (Summary)
3. Introduction
4. Main Content / Methodology
5. Results / Discussion
6. Conclusion
7. References.